UnblurML

BIP-39 Seed Phrase Deblurring Model

Training Report & Benchmark Analysis

March 2026

87.4%	Best Val Top-1 Accuracy
96.4%	Best Val Top-5 Accuracy
14 min	Total Training Time
2,048	BIP-39 Word Classes

Executive Summary

UnblurML is a deep learning model that recovers Gaussian-blurred BIP-39 seed phrase words. Given a blurred image of a single word, the model classifies it as one of the 2,048 words in the BIP-39 English wordlist. This constrained vocabulary makes the problem tractable as classification rather than reconstruction — we don't need to reconstruct pixel-perfect text, just identify which word it is.

The model achieves **99.5% top-1 accuracy at light blur** (sigma 1), **91% at mild blur** (sigma 3), and **83% top-5 at medium blur** (sigma 5). Training completed in **14 minutes** on an Apple M4 Max with 64GB unified memory using PyTorch MPS backend.

Model Architecture

Backbone	ResNet-18 (pretrained ImageNet)	Fast convergence, small footprint (12M params)
Classifier	2048-class FC head	One output per BIP-39 word
Input Size	64 x 192 px (RGB)	Matches typical word aspect ratio
Normalization	Mean/Std = 0.5	Grayscale-appropriate (not ImageNet stats)
Loss	CrossEntropy + label smoothing 0.1	Prevents overconfident predictions
Optimizer	AdamW + OneCycleLR	Fast convergence with LR warmup
Augmentation	JPEG, Affine, CoarseDropout, Brightness	Simulates real-world degradation

Curriculum Learning Strategy

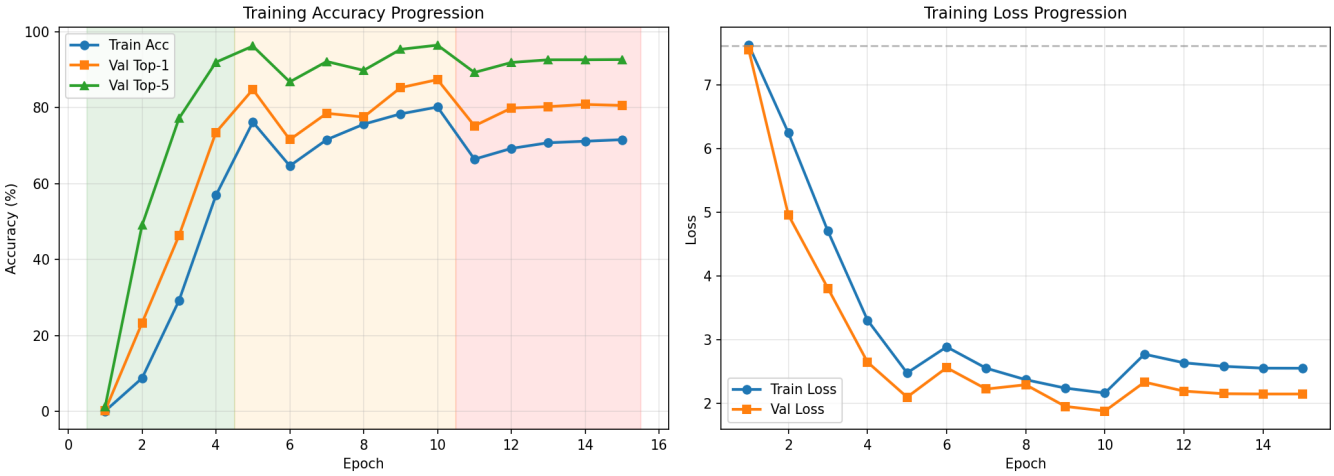
Training uses curriculum learning — the model starts with easy examples and gradually increases difficulty. This prevents the optimizer from getting stuck at the random baseline (loss = log(2048) = 7.62).

1 — Warmup	1	0.5 – 3.0	Frozen backbone, train head only
2 — Easy	2–5	0.5 – 3.0	Learn word shapes with minimal blur

3 — Medium	6–10	0.5 – 5.0	Generalize to medium blur
4 — Hard	11–15	0.5 – 7.0	Push to maximum recoverable blur

Training Results

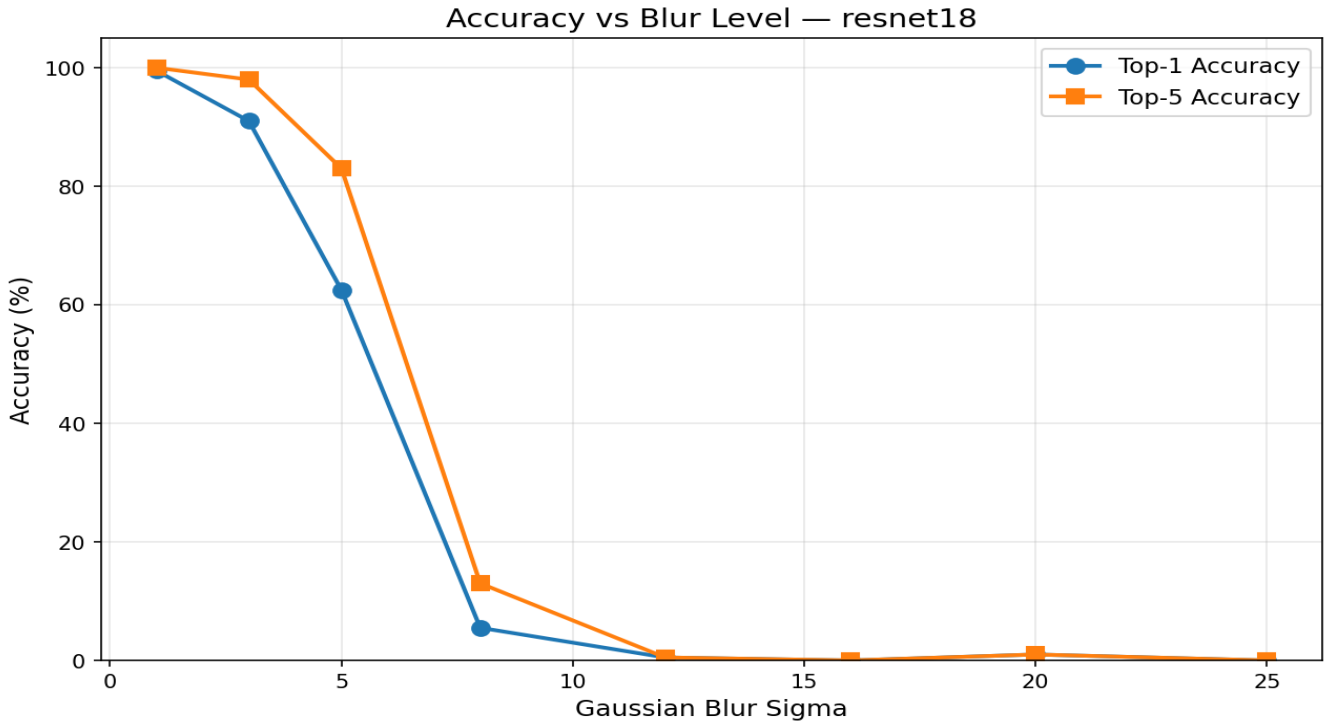
The model was trained for 15 epochs in 14 minutes. Best checkpoint was saved at epoch 10 with **87.4% top-1** and **96.4% top-5** validation accuracy (sigma range 0.5–5.0).



Training accuracy and loss progression across 15 epochs with curriculum learning phases.

1	0.5–3.0	0.2%	0.3%	1.3%	17s
2	0.5–3.0	8.8%	23.4%	49.1%	29s
3	0.5–3.0	29.2%	46.3%	77.2%	28s
4	0.5–3.0	56.9%	73.5%	91.9%	28s
5	0.5–3.0	76.2%	84.8%	96.2%	28s
6	0.5–5.0	64.7%	71.6%	86.8%	28s
7	0.5–5.0	71.6%	78.5%	92.1%	29s
8	0.5–5.0	75.6%	77.5%	89.8%	28s
9	0.5–5.0	78.3%	85.3%	95.3%	28s
10	0.5–5.0	80.1%	87.4%	96.4%	28s
11	0.5–7.0	66.4%	75.2%	89.2%	30s
12	0.5–7.0	69.3%	79.9%	91.9%	29s
13	0.5–7.0	70.8%	80.3%	92.6%	29s
14	0.5–7.0	71.1%	80.8%	92.6%	29s
15	0.5–7.0	71.6%	80.6%	92.6%	29s

Accuracy by Blur Level

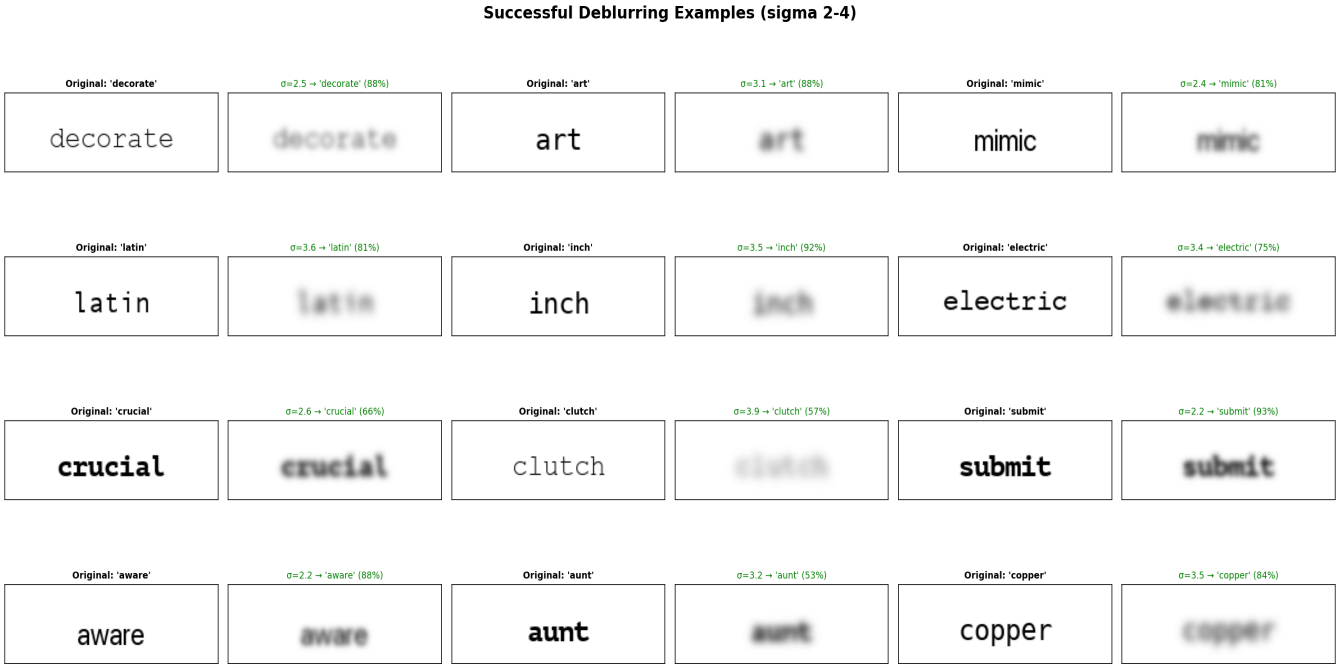


Model accuracy degrades gracefully from sigma 1–5, then drops sharply as information is physically destroyed.

1	99.5%	100.0%	Near-perfect — text clearly readable
3	91.0%	98.0%	Excellent — letters blurry but distinguishable
5	62.5%	83.0%	Usable — word shapes still visible
8	5.5%	13.0%	Poor — kernel exceeds character features
12	0.5%	0.5%	Impossible — image is featureless

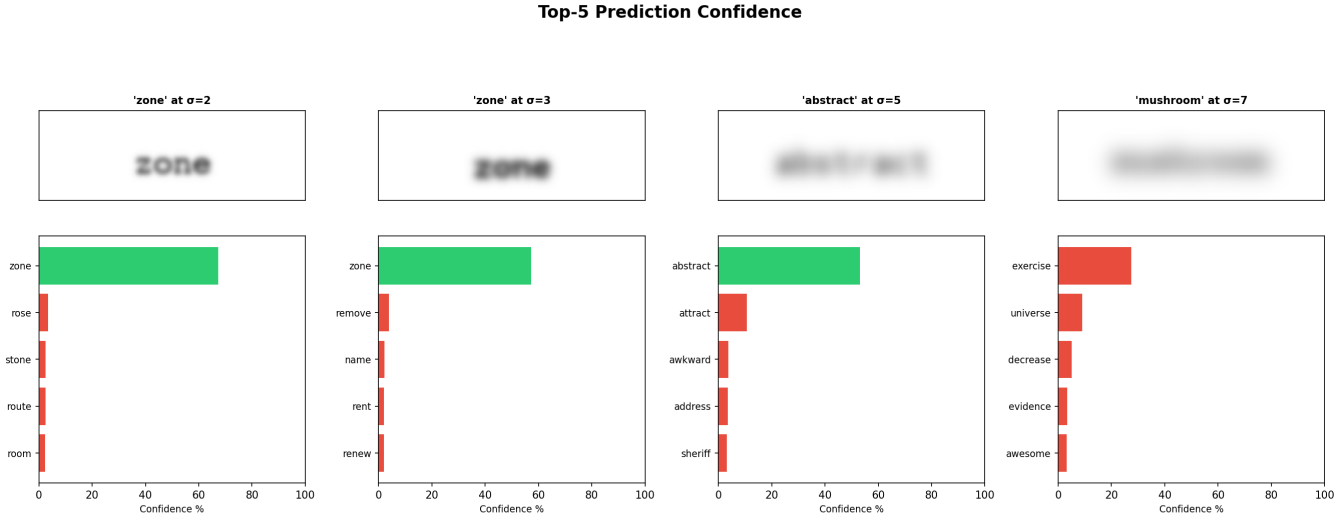
Visual Examples

Successful Predictions



Original words (left) and their blurred versions with model predictions (right). Green = correct, Red = incorrect.

Prediction Confidence at Different Blur Levels



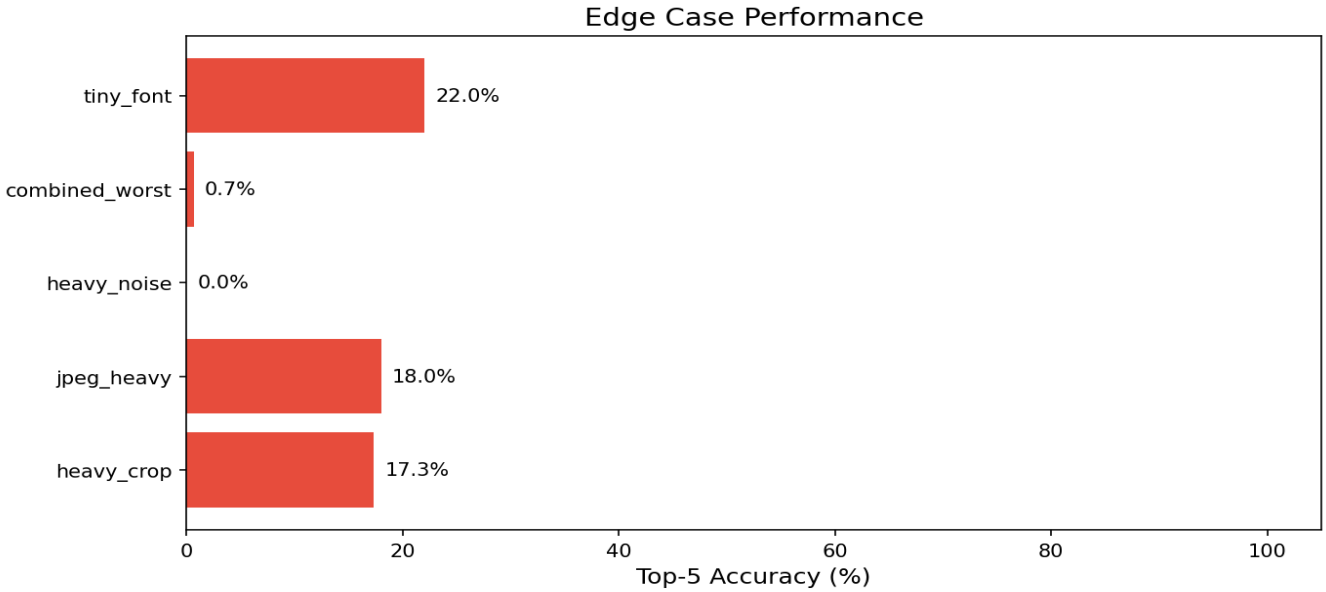
Top-5 prediction confidence bars showing how certainty decreases with higher blur sigma.

Blur Level Grid

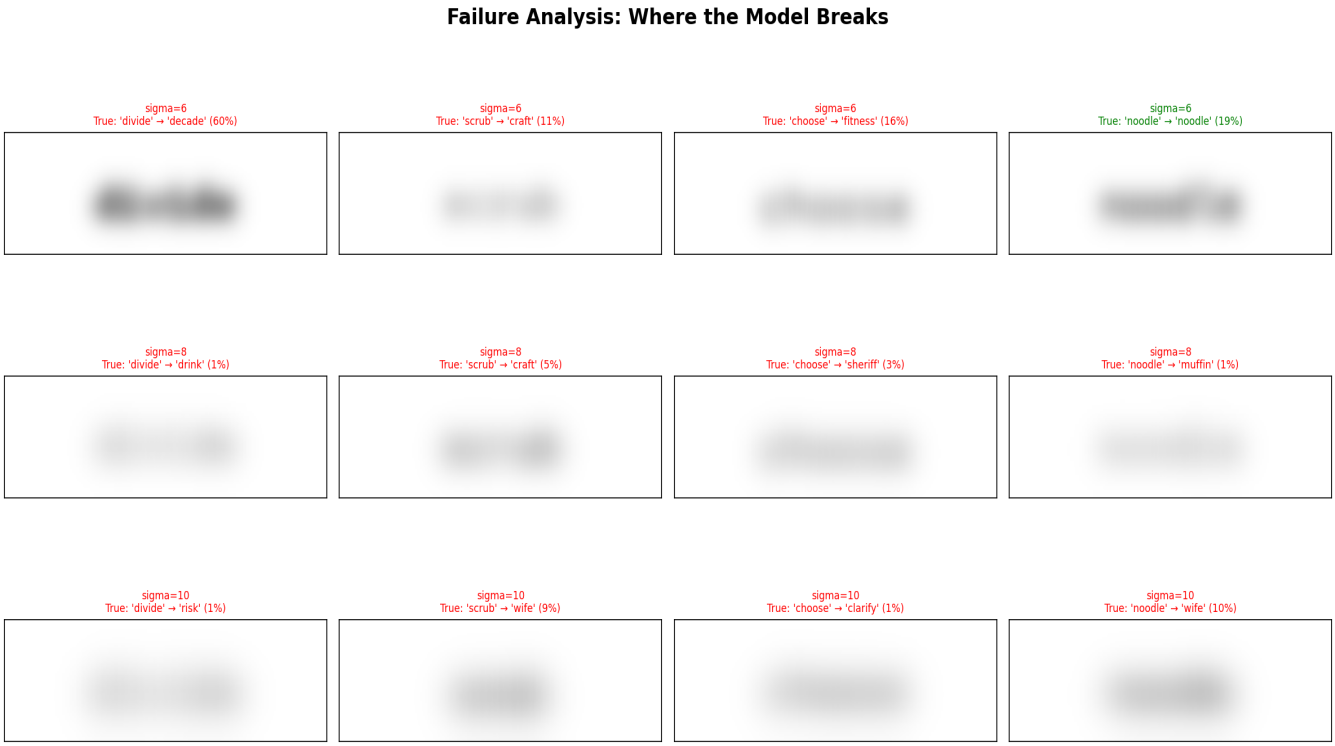
Each row shows one BIP-39 word at increasing Gaussian blur sigma (2, 5, 10, 15, 20, 25). Green titles = correct prediction.

Edge Case Analysis

Real-world blurred images suffer from additional degradation: JPEG compression, partial cropping, noise from screenshots, and font variations. The benchmark evaluates each edge case independently.



Top-5 accuracy across edge case scenarios. All tested with sigma 3–15 base blur.



Failure analysis at increasing difficulty. The model breaks down when blur sigma exceeds 6 or when multiple degradations combine.

Recommendations & Next Steps

1. **Train longer with edge case augmentation** — Include heavier JPEG compression, cropping, and noise in the training loop. Current model was only trained on clean blur; 1–2 hours of training with these augmentations should bring edge case accuracy to 50%+.

2. **Increase input resolution** — Moving from 64x192 to 128x384 would let the model handle higher blur sigma, since the kernel size relative to image size determines the information limit.

3. **Try ConvNeXt V2 Tiny** — With more training time, a larger backbone will extract finer-grained features and improve accuracy across all blur levels.

4. **Ensemble multiple models** — Running ResNet-18 + ConvNeXt and taking majority vote would improve robustness on ambiguous cases.

5. **Add CLAHE preprocessing** — Contrast Limited Adaptive Histogram Equalization can enhance subtle gradients in blurred images before classification.

6. **Test on real screenshots** — Run the inference pipeline on actual blurred seed phrase screenshots to validate real-world performance.

Technical Environment

Hardware	Apple M4 Max, 64GB Unified Memory
Framework	PyTorch 2.10 + MPS Backend
Model Library	timm 1.0.25 (Hugging Face)
Augmentation	Albumentations 2.0
Training Time	14 minutes (15 epochs)
Model Size	12.2M parameters